True Presence Detection — MOD/DOWA Fusion Engine

Technical Documentation v1.0

VDA · Telkonet — Group 7

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1. Overview

The fusion engine implements a **Multi-Sensor Occupancy Detection (MOD)** architecture using the **Data-driven Optimization-based Weighted Average (DOWA)** method, with evidence combination based on **Dempster-Shafer Theory (DST)**. The engine receives raw telemetry from five sensor modalities, produces a normalized evidence mass for each, combines them into a single occupancy probability, and outputs a binary occupancy state: **OCCUPIED (1)** or **EMPTY (0)**.

The system is designed to run entirely on the Local Brain (ESP32 central microcontroller), with no raw sensor data transmitted beyond the room boundary. Only the final binary output is published to the MQTT broker.

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2. Sensor Stack and Roles

The system integrates five sensor modalities organized across two physical units:

2.1 PIR Sensor (Ceiling-mounted)

- **Physical principle:** Detects changes in infrared radiation caused by transversal body movement.
- **Role in fusion:** Fast-response motion trigger. Primary detector for active presence (walking, entering, exiting).
- **Fundamental limitation:** Blind to stationary or sleeping occupants. Zero signal during sleep.
- **Fusion classification:** High-speed, low-persistence signal.

2.2 mmWave Radar LD2410 24GHz (Bajoore Hub)

- **Physical principle:** Emits millimeter-wave pulses and analyzes reflected Doppler signatures.
- **Role in fusion:** Stationary presence detector. Captures micro-movements including chest expansion from breathing (~0.3–0.5 cm displacement) and heartbeat vibration (~0.01 cm displacement).
- **Fundamental limitation:** Susceptible to clutter from structural elements (pipes, water). Mitigated by RF-absorber backing.
- **Fusion classification:** Medium-speed, high-persistence signal. Most critical sensor during sleep.

2.3 CO_2 Sensor — Sensirion SCD41 (Bajoore Hub)

- **Physical principle:** NDIR optical measurement of CO_2 concentration in ambient air.
- **Role in fusion:** Slow but motion-independent presence confirmation. CO_2 accumulates steadily from human respiration regardless of movement.
- **Fundamental limitation:** Inherent time lag of 60–180 seconds. Cannot detect instantaneous presence changes.
- **Fusion classification:** Slow, high-persistence, high-confidence signal for prolonged occupancy.

2.4 Temperature Sensor — Sensirion SHT41 (Bajoore Hub)

- **Physical principle:** Capacitive measurement of ambient temperature.
- **Role in fusion:** Weak reinforcement signal. Human body heat marginally raises ambient temperature over time.
- **Fundamental limitation:** Very slow response (minutes to hours). Affected by HVAC system and seasonal variations. Unreliable as standalone signal.
- **Fusion classification:** Very slow, low-weight confirmatory signal.

2.5 Humidity Sensor — Sensirion SHT41 (Bajoore Hub)

- **Physical principle:** Capacitive measurement of relative humidity.
- **Role in fusion:** Weak reinforcement signal. Human respiration marginally raises ambient humidity over time.
- **Fundamental limitation:** Same as temperature — very slow, easily confounded by environmental factors.
- **Fusion classification:** Very slow, low-weight confirmatory signal.

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3. Theoretical Framework: Dempster-Shafer Theory

3.1 Frame of Discernment

The system operates on a Frame of Discernment Θ with three mutually exclusive hypotheses:

- **O** Room is OCCUPIED
- **E** Room is EMPTY
- **U** UNKNOWN (sensor cannot determine state)

Every sensor produces a **Basic Probability Assignment (BPA)** — commonly called a mass function — that distributes belief across these three hypotheses. For each sensor *i*:

- **m_i(O)** mass assigned to OCCUPIED hypothesis
- **m_i(E)** mass assigned to EMPTY hypothesis
- **m_i(U)** mass assigned to UNKNOWN (sensor uncertainty)

The constraint is: **m_i(O) + m_i(E) + m_i(U) = 1.0** for every sensor at every timestep.

This is the fundamental advantage of Dempster-Shafer over simple Bayesian weighting: a sensor that cannot determine the state — for example, CO_2 in the first 90 seconds after a guest enters — explicitly contributes high mass to UNKNOWN rather than distorting the fusion with a false signal.

3.2 DOWA Weight Assignment

In the DOWA method, each sensor is assigned a **confidence weight w_i** that reflects its expected detection reliability for binary occupancy in a hotel room context. These weights are initialized from published literature accuracy figures and will be refined during prototype calibration.

The weights are **not uniform across time of day** — the system applies contextual weight modulation based on temporal context (day mode vs. night mode), as occupancy patterns in a hotel room differ significantly between active hours and sleeping hours.

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4. Mass Function Computation per Sensor

This section defines how each sensor's raw reading is converted into a Dempster-Shafer mass function at each measurement cycle (default: every 5 seconds).

4.1 PIR Sensor — Mass Function

Base weight: w_PIR = 0.25 (day mode) / 0.10 (night mode 23:00–07:00)

The PIR output is binary. However, rather than mapping directly to m(O) = 1 or m(E) = 1, a **hold-decay** mechanism is applied to avoid mass discontinuities:

Condition	m(O)	m(E)	m(U)
Active trigger (motion detected)	0.90	0.00	0.10
Hold period (0–30s after last trigger)	0.75	0.00	0.25
Decay period (30–120s after last trigger)	linear decay from 0.75 to 0.00	0.00	linear rise to 1.00
No trigger (>120s silence)	0.00	0.30	0.70

Rationale: When PIR detects no motion for >120 seconds, it cannot conclude the room is empty — the guest may be sleeping. Therefore m(E) remains low (0.30) and most mass flows to UNKNOWN. This is the key difference from a naive rule-based system, which would classify no-motion as empty.

Night mode rationale: During 23:00–07:00, the PIR base weight is reduced to 0.10 because a sleeping guest produces zero motion signal for hours. Assigning high weight to PIR at night would systematically produce false EMPTY classifications.

4.2 mmWave LD2410 — Mass Function

Base weight: w_mmWave = 0.35 (day mode) / 0.50 (night mode)

The LD2410 provides three native states: Moving, Stationary, None. Each maps to a distinct mass function:

LD2410 State	m(O)	m(E)	m(U)
Moving (gross movement detected)	0.95	0.00	0.05
Stationary (micro-movement / breathing detected)	0.85	0.00	0.15
None (no signal)	0.05	0.70	0.25

Rationale for Stationary state: The LD2410 in stationary mode detects chest expansion from respiration (~0.3–0.5 cm) and heartbeat vibration (~0.01 cm). This is the primary mechanism for sleeping guest detection. The mass assignment m(O) = 0.85 is deliberately high but not 1.0, because the sensor can occasionally detect false stationary signals from vibration sources (HVAC, structural). The RF-absorber backing mitigates this but does not eliminate it entirely.

Rationale for None state: When the LD2410 returns None, it assigns m(E) = 0.70 rather than 1.0, because the sensor has a physical detection cone limited by its ±60° angle and 5m range. A guest standing outside the cone (e.g., in the bathroom) would return None while the room is occupied.

Night mode rationale: w_mmWave rises to 0.50 at night because it becomes the single most reliable sensor for sleeping presence detection. CO_2 and temperature/humidity provide slow confirmatory support; mmWave is the primary evidence source.

4.3 CO_2 Sensor SCD41 — Mass Function

Base weight: w_CO2 = 0.20 (day mode) / 0.25 (night mode)

CO_2 mass functions are computed using a **sigmoid transformation** applied to the delta between current reading and the room baseline:

Baseline: CO_2 concentration measured during confirmed empty room periods. Typical hotel room baseline: 420–480 ppm. Updated dynamically during confirmed EMPTY periods.

Delta: $\Delta = CO_2\_current - CO_2\_baseline$

Δ (ppm)	m(O)	m(E)	m(U)
$\Delta < 50$	0.00	0.60	0.40
$50 \le \Delta < 150$	0.20	0.30	0.50
$150 \le \Delta < 300$	0.55	0.10	0.35
$300 \le \Delta < 500$	0.80	0.00	0.20
$\Delta \ge 500$	0.92	0.00	0.08

Critical temporal modifier: CO_2 accumulates slowly. For the first 180 seconds after any occupancy state change, the CO_2 mass function is attenuated by a **lag penalty factor of**

0.5 applied to m(O) and m(E), with the remainder redistributed to m(U). This prevents CO_2 from contributing false evidence during its characteristic accumulation delay.

Night mode rationale: w_CO2 increases to 0.25 at night because a sleeping guest continuously produces CO_2 for hours. After 2+ hours of sleep, Δ values of 300–500 ppm are typical, producing strong confirmatory evidence for the mmWave reading.

4.4 Temperature Sensor SHT41 — Mass Function

Base weight: w_temp = 0.10 (day mode) / 0.08 (night mode)

Temperature mass functions use delta from baseline, measured in °C:

Baseline: Average temperature during confirmed empty periods in the same room, updated daily.

ΔT (°C)	m(O)	m(E)	m(U)
ΔT < 0.3	0.00	0.30	0.70
0.3 ≤ ΔT < 0.7	0.20	0.10	0.70
0.7 ≤ ΔT < 1.2	0.45	0.00	0.55
ΔT ≥ 1.2	0.65	0.00	0.35

Rationale: Temperature receives the second-lowest base weight because it is highly confounded by the HVAC system, window state, and time of day. The mass function deliberately keeps m(U) high across all ranges to reflect this uncertainty. Even at ΔT ≥ 1.2°C — an unusually strong human thermal signature — m(O) is capped at 0.65 rather than approaching certainty.

Night mode rationale: Slightly reduced weight at night because HVAC systems often switch to night setpoint, introducing systematic temperature drift unrelated to occupancy.

4.5 Humidity Sensor SHT41 — Mass Function

Base weight: w_hum = 0.10 (day mode) / 0.07 (night mode)

Humidity mass functions use delta from baseline in %RH:

ΔRH (%)	m(O)	m(E)	m(U)

ΔRH < 1.5	0.00	0.25	0.75
1.5 ≤ ΔRH < 3.0	0.15	0.10	0.75
3.0 ≤ ΔRH < 6.0	0.40	0.00	0.60
ΔRH ≥ 6.0	0.60	0.00	0.40

Rationale: Humidity receives the lowest base weight. Ambient humidity is highly affected by shower use, window opening, and external weather conditions — all unrelated to presence. The mass function is intentionally conservative, with m(U) dominant across most ranges.

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5. DOWA Combination Procedure

After computing the mass functions for all active sensors, the DOWA procedure combines them into a single fused mass function using the **Dempster-Shafer combination rule**.

5.1 Weight Normalization

At each cycle, the active sensor weights are collected and normalized to sum to 1.0:

For N active sensors with base weights w_1, w_2, ..., w_N (adjusted for time-of-day context):

w_i_normalized = w_i / Σ(w_j) for j = 1 to N

A sensor is considered **inactive** and excluded from normalization if:

- Its last reading timestamp exceeds 60 seconds (communication failure)
- It reports a hardware error flag
- It is in its lag period (for CO_2: first 180s after state change)

5.2 Weighted Mass Aggregation

The DOWA method first produces a **weighted aggregate mass** for each hypothesis before applying Dempster-Shafer combination:

For each hypothesis H {O, E, U}:

m_DOWA(H) = Σ [w_i_normalized × m_i(H)] for i = 1 to N

This produces an intermediate mass function that already reflects sensor reliability weighting.

5.3 Dempster-Shafer Combination

The Dempster-Shafer combination rule is then applied to combine the DOWA-weighted evidence from independent sensor pairs. The combination of two mass functions m_A and m_B produces:

m_combined(H) = [Σ m_A(X) × m_B(Y) for all X Y=H] / (1 − K)

Where K is the **conflict coefficient**:

K = Σ m_A(X) × m_B(Y) for all X Y=

K measures how much two sensors contradict each other. High K (approaching 1.0) signals sensor conflict and triggers a **conflict alert** — logged for diagnostic purposes, with the fused mass defaulting to higher UNKNOWN mass until resolved.

The combination is applied sequentially across all N sensors, producing a final fused mass function:

m_fused(O), m_fused(E), m_fused(U)

5.4 Decision Rule

The final binary occupancy decision is made by comparing m_fused(O) against a threshold:

Condition	Output
m_fused(O) ≥ 0.50	**OCCUPIED (1)**
m_fused(O) < 0.50	**EMPTY (0)**

A **hysteresis mechanism** is applied to prevent rapid state toggling:

- Transition from EMPTY OCCUPIED requires m_fused(O) ≥ 0.50 for **2 consecutive cycles** (10 seconds)
- Transition from OCCUPIED EMPTY requires m_fused(O) < 0.50 for **6 consecutive cycles** (30 seconds)

The asymmetric hysteresis is intentional: it is worse to incorrectly declare EMPTY (cutting HVAC on a sleeping guest) than to maintain OCCUPIED for 30 extra seconds after departure.

6. Contextual Weight Modulation

6.1 Day Mode vs. Night Mode

Sensor	Day weight	Night weight (23:00–07:00)
PIR	0.25	0.10
mmWave	0.35	0.50
CO_2	0.20	0.25
Temperature	0.10	0.08
Humidity	0.10	0.07
Total	**1.00**	**1.00**

6.2 Dynamic Weight Degradation

During runtime, individual sensor weights are dynamically degraded under the following conditions:

Condition	Action
Sensor reading age > 60s	w_i × 0.5
Sensor hardware error flag	w_i = 0.0 (excluded)
CO_2 in lag period (first 180s)	w_CO2 × 0.5, extra mass U
PIR consecutive triggers > 50 in 60s	w_PIR × 0.7 (possible false positive source)
mmWave conflict with CO_2 for > 300s	conflict alert logged, w_U increased

6.3 BLE Staff Tag Override

When the ESP32 receiver detects a valid BLE staff tag (RFID/BLE beacon with authorized UUID) within RSSI threshold for room-level proximity:

- The fusion engine **freezes the current occupancy state output**
- A **STAFF_PRESENT flag** is set to TRUE

- The binary output remains unchanged from the pre-staff-entry value
- Staff presence is logged separately for housekeeping workflow records
- Upon tag departure (RSSI below threshold for 10 consecutive seconds), the STAFF_PRESENT flag clears and the fusion engine resumes normal operation

This ensures that housekeeping activity does not alter guest occupancy state records.

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7. Worked Example: Sleeping Guest at 03:00

Context: Guest entered room at 22:30, has been sleeping since 23:15. Current time: 03:00. Night mode active.

Raw sensor readings:

- PIR: No trigger for 4 hours hold expired, no motion
- mmWave: State = Stationary (breathing signature detected)
- CO_2: Current = 820 ppm, Baseline = 450 ppm Δ = 370 ppm
- Temperature: Current = 22.8°C, Baseline = 21.5°C ΔT = 1.3°C
- Humidity: Current = 54%, Baseline = 49% ΔRH = 5.0%

Mass functions computed (night mode weights):

Sensor	w_i	m(O)	m(E)	m(U)
PIR	0.10	0.00	0.30	0.70
mmWave	0.50	0.85	0.00	0.15
CO_2	0.25	0.80	0.00	0.20
Temperature	0.08	0.65	0.00	0.35
Humidity	0.07	0.40	0.00	0.60

DOWA weighted aggregate:

- m_DOWA(O) = (0.10×0.00) + (0.50×0.85) + (0.25×0.80) + (0.08×0.65) + (0.07×0.40) = 0.000 + 0.425 + 0.200 + 0.052 + 0.028 = **0.705**

- m_DOWA(E) = (0.10×0.30) + (0.50×0.00) + (0.25×0.00) + (0.08×0.00) + (0.07×0.00) = 0.030 + 0.000 + 0.000 + 0.000 + 0.000 = **0.030**

- m_DOWA(U) = 1.00 − 0.705 − 0.030 = **0.265**

After Dempster-Shafer normalization (K is low, conflict minimal):

- m_fused(O) ≈ **0.731**
- m_fused(E) ≈ **0.031**
- m_fused(U) ≈ **0.238**

Decision: m_fused(O) = 0.731 ≥ 0.50 **OUTPUT: OCCUPIED (1)**

The system correctly identifies the sleeping guest despite PIR returning zero signal for 4 hours.

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8. Known Limitations and Mitigation

Limitation	Impact	Mitigation
CO_2 time lag (60–180s)	Slow response to room entry	PIR and mmWave provide fast initial detection; CO_2 reinforces after lag
mmWave clutter from pipes/water	False stationary readings	RF-absorber backing; clutter filtering via LD2410 firmware thresholds
Temperature/humidity confounded by HVAC	Unreliable standalone signal	Low weight (0.08–0.10); high m(U) mass; confirmatory role only
PIR blind to stationary occupants	False EMPTY at night	Night mode reduces PIR weight to 0.10; mmWave takes primary role
BLE RSSI multipath in hotel rooms	False staff detection	RSSI threshold set conservatively; 10-second persistence required
Dempster-Shafer conflict (K 1.0)	Unreliable fusion output	Conflict alert logged; output defaults to OCCUPIED (fail-safe)

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9. Summary of Design Principles

1. **No single sensor is trusted absolutely.** Every sensor contributes probabilistic evidence, not binary truth.
2. **Uncertainty is explicit.** The UNKNOWN mass class ensures that sensor silence is not misinterpreted as absence.

3. **The cost of false EMPTY exceeds the cost of false OCCUPIED.** All design choices — hysteresis asymmetry, fail-safe on conflict, conservative PIR decay — reflect this asymmetry.
4. **Context modulates confidence.** Day and night modes reflect the physical reality of hotel room occupancy patterns.
5. **All processing is local.** Only the final binary output crosses the network. No raw sensor data leaves the room.